B.TECH CAPSTONE PROJECT DEFENSE // ARTIFICIAL INTELLIGENCE & DATA ANALYTICS 2026

TRUTHGUARD

An Explainable Framework for Hallucination Detection, Evidence Verification, and Evidence-Grounded Automatic Correction in LLM Responses

A claim-level natural language processing framework providing token-level provenance, calibrated cross-encoder arbitration, and closed-loop self-correction.

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SPECIFICATION
Research Architecture
Hybrid BM25 + FAISS Vector Index
DeBERTa-v3 NLI + Closed-Loop Audit

Abstract

Addressing ungrounded language model generation through fine-grained verification and self-correction.

01. THE PROBLEM

Large Language Models routinely produce fluent, grammatically flawless outputs that are factually ungrounded ('hallucinations'). In safety-critical fields such as medicine, law, and engineering, these undetectable errors present severe risks of misinformation and silent automated failures.

02. THE APPROACH

TruthGuard replaces opaque, whole-answer prompt grading with fine-grained syntactic atomic claim decomposition, character-span tracking, and hybrid dense-sparse evidence retrieval. Each proposition is verified independently against indexed reference documents.

03. CORE COMPONENTS

The framework integrates sentence-transformers (all-MiniLM-L6-v2), exact cosine similarity in FAISS IndexFlatIP (384-d), Okapi BM25 keyword matching, and a DeBERTa-v3 cross-encoder NLI model with joint premise-hypothesis attention.

04. INTENDED OUTCOME

The system delivers auditable document citations, separates model uncertainty from missing evidence, and executes closed-loop reverification to ensure rewritten answers eliminate secondary hallucinations with zero remaining contradictions.

Problem Statement: Fluency is Not Factuality

Autoregressive token prediction optimizes for surface likelihood rather than empirical truth.

THE ILLUSION: SURFACE FLUENCY

"The James Webb Space Telescope took the very first direct pictures of an exoplanet in 1995, revealing Kepler-186f."

- Linguistic Perplexity: Low: Grammatically flawless, authoritative syntax, and persuasive tone.
- Statistical Likelihood: High probability under standard autoregressive token sampling $P(w_t \mid w_{<t})$.
- The Cognitive Deception: Evaluators and prompt critics routinely mistake fluency for truth.

EMPIRICAL REALITY: FACTUAL CONTRADICTIONS

Verified Ground Truth Reference Facts:

✕ Temporal Conflict: JWST launched on December 25, 2021; scientific imaging began mid-2022.
✕ Historical Conflict: The first direct exoplanet photograph was 2M1207b in 2004 by the ESO VLT.
✕ Astronomical Conflict: Kepler-186f was detected via transit photometry by the Kepler telescope in 2014.
✕ Architectural Cause: Autoregressive models have no intrinsic grounding or self-audit mechanism.

THE WHOLE-ANSWER ASSESSMENT TRAP

Grading entire paragraphs at once obscures individual false clauses behind mostly true context. An answer can be 90% accurate while containing one fatal factual error. True reliability demands atomic, claim-level evidence verification.

Research Objectives & Critical Gaps

Addressing the fundamental limitations of conventional verification approaches.

CONVENTIONAL VERIFICATION LIMITATIONS

✕ **Monolithic LLM-as-a-Judge:**
Prompting another LLM to score full answers is opaque, expensive, and subject to second-order hallucinations without verifiable citations.

✕ **Pure Dense Semantic Retrieval:**
Cosine similarity alone often misses exact named entities, specific model codes, and numerical dates without keyword matching.

✕ **Bi-Encoder Verification Inadequacy:**
Bi-encoders encode claim and evidence separately, failing to detect subtle logical negations (e.g. 'did not win' vs 'won').

✕ **Unconstrained Open-Loop Editing:**
Standard prompt-based corrections frequently introduce secondary hallucinations while attempting to edit earlier mistakes.

TRUTHGUARD RESEARCH OBJECTIVES

✓ **Atomic Syntactic Decomposition:**
Isolate testable empirical propositions with exact character-span offsets [start, end] matching the original string.

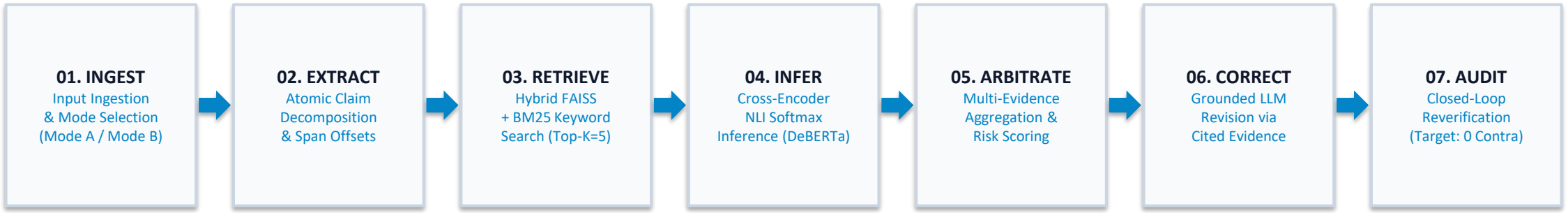
✓ **Hybrid FAISS + BM25 Retrieval:**
Combine 384-d dense semantic embeddings with sparse Okapi BM25 keyword matching ($\alpha = 0.65$) to maximize retrieval recall.

✓ **Cross-Encoder Softmax Arbitration:**
Execute joint premise-hypothesis attention using DeBERTa-v3 with calibrated decision boundaries (0.55 / 0.50).

✓ **Closed-Loop Reverification Audit:**
Constrain LLM revision to verified reference facts and mandate a second verification pass to guarantee 0 remaining errors.

Proposed Methodology: End-to-End Workflow

A sequential, mathematically defined verification and self-correction pipeline.



PHASE I: DECOMPOSITION & EVIDENCE RETRIEVAL

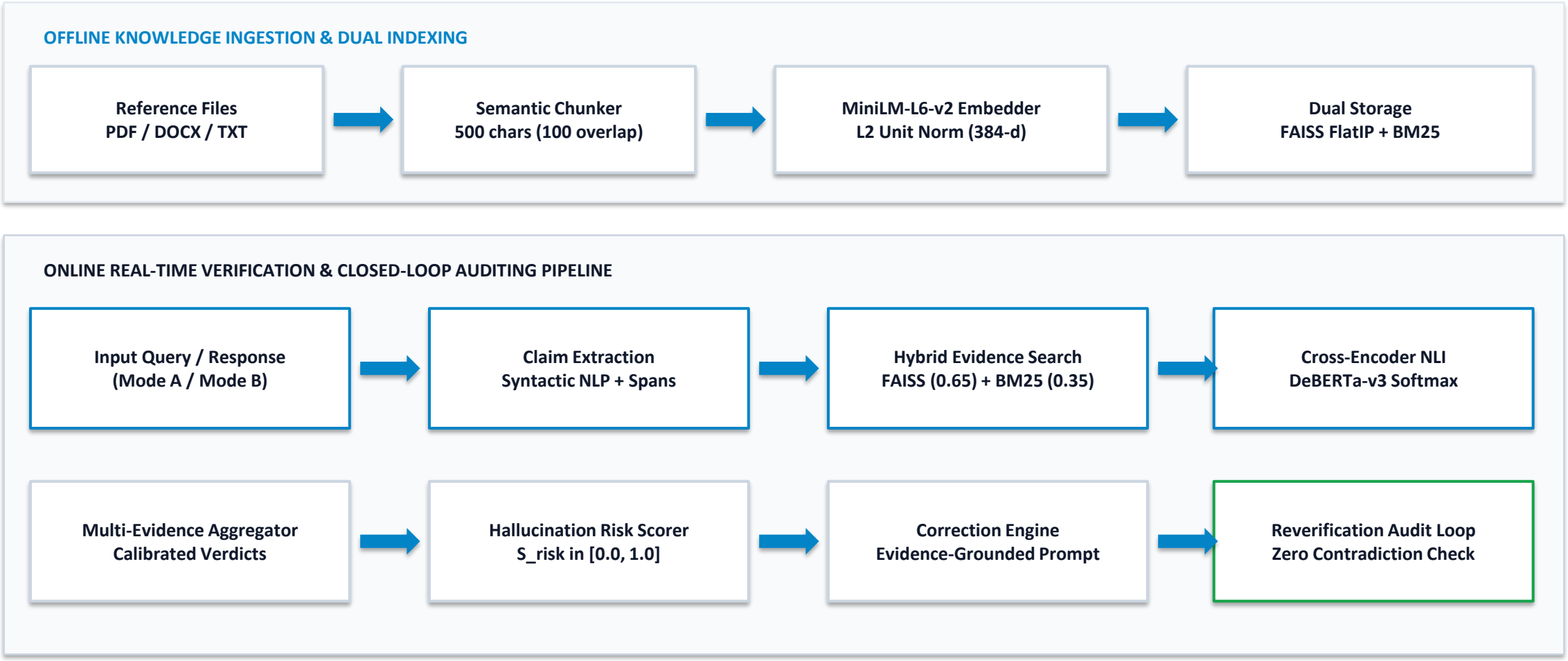
- Atomic Extraction: Splits complex text into self-contained propositions, resolving anaphoric pronouns (he/she/it).
- Character Span Tracking: Logs exact start and end string offsets to preserve token provenance.
- Dual Vector & Keyword Index: Encodes queries via MiniLM-L6-v2 and searches FAISS IndexFlatIP alongside BM25.

PHASE II: INFERENCE, CORRECTION & REVERIFICATION

- Cross-Encoder Arbitration: Concatenates evidence premise and claim hypothesis for joint token attention in DeBERTa-v3.
- Calibrated Confidence: Applies empirical thresholds (0.55 entailment, 0.50 contradiction) and logs diagnostic states.
- Reverification Audit Loop: Rewritten responses undergo full re-verification; output is delivered only when contradictions equal zero.

System Architecture & Component Boundaries

Modular architecture separating offline knowledge indexing from real-time online verification.



Atomic Claim Decomposition & Character Span Tracking

Deconstructing compound assertions into standalone testable propositions with exact string offsets.

EXAMPLE INPUT COMPOUND SENTENCE (EXACT STRING LENGTH: 76 CHARACTERS)

"Marie Curie discovered radium in 1898 and won the Nobel Prize in Literature."

`[Span 0:37: Factual Discovery Claim]` `[Span 42:75: Fabricated Award Claim]`

Claim ID	Original Substring Span	Normalized Standalone Proposition	Char Span	Final Verdict
C-01	"Marie Curie discovered radium in 1898"	"Marie Curie discovered radium in 1898."	[0 : 37]	SUPPORTED
C-02	"won the Nobel Prize in Literature"	"Marie Curie won the Nobel Prize in Literature."	[42 : 75]	CONTRADICTED

DEPENDENCY PARSING

Identifies coordinating conjunctions ('and', 'but', 'whereas') to split compound sentences into standalone grammatical clauses without semantic loss.

ANAPHORA RESOLUTION

Resolves anaphoric pronouns ('she' —► 'Marie Curie') so each extracted claim is fully self-contained for isolated retrieval and verification.

PINPOINT SPAN MAPPING

Preserves exact [start_offset, end_offset] character positions. The UI highlights only the false phrase; verified facts remain completely untouched.

Evidence Retrieval Using FAISS & BM25

Fusing FAISS inner product cosine similarity with Okapi BM25 keyword inverted indexing.

MATHEMATICAL RETRIEVAL PIPELINE

- **Text Chunking:**
  Documents partitioned into 500-character passages with 100-character overlap to preserve cross-boundary semantic context.
- **Dense Embedding Generation:**
  MiniLM-L6-v2 maps text to 384-dimensional dense vectors: v in R^384. Vectors are strictly L2-normalized: v_hat = v / ||v||_2.
- **FAISS Vector Indexing:**
  faiss.IndexFlatIP(384) computes exact inner products: sim(q, v) = q_hat · v_hat = cos(θ). Zero approximation error.
- **Sparse Lexical Indexing:**
  Okapi BM25 inverted index (k1 = 1.5, b = 0.75) catches exact entity tokens ('Marie Curie', 'radium', '1898').
- **Convex Score Fusion:**
  S_hybrid = 0.65 · S_dense + 0.35 · S_BM25. Prunes candidates with similarity below threshold τ = 0.25.

EVIDENCE PROVENANCE & DIAGNOSTIC TRACKING

► **Top-K Candidate Selection:**
Retrieves top K = 5 candidate passages meeting the composite relevance threshold for downstream cross-encoder evaluation.

► **Explicit Source Provenance:**
Every passage record maintains metadata: document_id, document_name, page_number, chunk_id, and raw text.

► **Empty Retrieval Diagnosis:**
Separates three distinct operational causes when 0 passages are returned: NO_DOCUMENTS_INDEXED, NO_DOCUMENTS_SELECTED, NO_EVIDENCE_RETRIEVED.

► **Hardware Preservation:**
Runs embeddings and FAISS index entirely on multi-threaded CPU memory, reserving dedicated GPU VRAM for LLM generation.

NLI-Based Claim Verification & Softmax Arbitration

Joint premise-hypothesis cross-attention and multi-passage verdict arbitration.

CROSS-ENCODER JOINT ATTENTION

- **Model Selection:**
  cross-encoder/nli-deberta-v3-small (512 max token sequence length).
- **Joint Sequence Input:**
  Concatenates input as: [CLS] Evidence Premise P [SEP] Claim Hypothesis H [SEP]. All tokens attend to each other simultaneously.
- **Why Bi-Encoders Fail:**
  Bi-encoders encode P and H separately, missing subtle logical negations (e.g. 'did not win' vs 'won') that cross-encoders catch.
- **Softmax Probability Logits:**
  P(label_k) = exp(z_k) / sum(exp(z_j)) across three classes: Contradiction (Index 0), Entailment (Index 1), Neutral (Index 2).
- **Multi-Passage Aggregation:**
  Synthesizes up to 5 passage-level predictions into a single claim-level verdict using formal arbitration rules.

DECISION BOUNDARIES & CALIBRATED CONFIDENCE

► **Supported Decision Rule:**
P(entailment) >= 0.55 and P(entailment) > P(contradiction) —► Verdict: Supported (Confidence: P(entail)).

► **Contradicted Decision Rule:**
P(contradiction) >= 0.50 and P(contradiction) > P(entailment) —► Verdict: Contradicted (Confidence: P(contra)).

► **Conflicting Evidence Rule:**
Distinct passages produce both entailment and contradiction within margin 0.15 —► Verdict: Conflicting Evidence.

► **Insufficient Evidence Rule:**
Max probabilities fail to meet thresholds or NLI classifies neutral —► Verdict: Insufficient Evidence.

► **Calibration Integrity:**
TruthGuard flags confidence_calibrated = True only when backed by genuine NLI probability, separating model uncertainty from missing data.

Evidence-Grounded Correction and Re-verification

Constrained text regeneration paired with mandatory post-edit reverification to eliminate secondary hallucinations.

STEP 1: AUDIT REPORT
Verification Service identifies Supported vs Contradicted claims and attaches refuting passages.

STEP 2: REVISION PROMPT
Prompt enforces verbatim retention of verified facts & strict evidence substitution.

STEP 3: LOCAL REVISION
Ollama Qwen-3:8B (temp 0.1) rewrites false claims strictly using cited reference passages.

STEP 4: REVERIFICATION
Corrected response is re-fed through TruthGuard to verify 0 remaining contradictions.

TRACED END-TO-END REVISION & REVERIFICATION WALKTHROUGH

- **Original Generated Answer:**
  "The Apollo 11 mission landed on the Moon in 1969, led by Commander Yuri Gagarin."
- **Initial Verification Verdicts:**
  Claim 1: Landing in 1969 —► SUPPORTED (P(entail) = 0.96) | Claim 2: Led by Gagarin —► CONTRADICTED (P(contra) = 0.91)
- **Retrieved Ground Truth Passage:**
  "NASA Records (p. 1): Neil Armstrong served as Commander of Apollo 11; Yuri Gagarin piloted Vostok 1 in 1961."
- **Evidence-Grounded Revised Answer:**
  "The Apollo 11 mission landed on the Moon in 1969, led by Commander Neil Armstrong [NASA-Records, p. 1]."
- **Mandatory Reverification Audit:**
  Re-extracted claims: 'Apollo 11 landed in 1969' —► SUPPORTED; 'Apollo 11 led by Armstrong' —► SUPPORTED. Remaining contradictions: 0. Certified faithful.

Datasets & Production Technology Stack

Empirical benchmark corpora and optimized local engineering architecture.

BENCHMARK DATASETS & EVALUATION ROLE

- **FEVER Benchmark:**
  Fact Extraction and VERification dataset. Sentence-level claims paired with Wikipedia reference corpus. Label space: SUPPORTS, REFUTES, NOT ENOUGH INFO. Evaluates isolated retrieval Recall@K and NLI accuracy on paper_dev split.
- **FEVER Repository Files:**
  datasets/fever/raw/paper_dev.jsonl (2.1 MB), datasets/fever/prepared/fever_paper_dev_1000.jsonl (476 KB), and 30+ cached Wikipedia articles in datasets/fever/evidence_cache/.
- **HaluEval Benchmark:**
  Hallucination Evaluation for Large Language Models. Evaluates full-length QA and summarization generations labeled as Faithful vs Hallucinated. Tests end-to-end atomic decomposition and risk scoring.
- **Zero Data Leakage Protocol:**
  Gold benchmark labels are strictly withheld from the prediction pipeline; the framework operates purely as an extrinsic evaluator with zero prior exposure to test queries.

IMPLEMENTED TECHNOLOGY STACK

- **API & Core Architecture:**
  FastAPI (Python 3.11), Pydantic v2 schemas, SQLAlchemy ORM, SQLite audit database.
- **NLP & Embedding Models:**
  SentenceTransformers (all-MiniLM-L6-v2, 384-d), Hugging Face Cross-Encoder (nli-deberta-v3-small).
- **Vector & Keyword Indices:**
  FAISS IndexFlatIP (exact cosine similarity), In-memory persistent Okapi BM25 index.
- **Local LLM Generation:**
  Ollama runtime (qwen3:8b default, qwen2.5:3b fallback), temperature 0.1, 1024 max tokens.
- **Hardware & VRAM Strategy:**
  RTX 3050 Laptop GPU (4GB VRAM) dedicated to Ollama; NLI and embeddings run on CPU threads to eliminate OOM thrashing.
- **Frontend Dashboard:**
  React 18, Vite, Tailwind CSS, Lucide icons, interactive claim span highlighter.

Experimental Results & Performance Metrics

Empirical confusion matrix, classification metrics, and warm latency breakdown.

90.0%
OVERALL ACCURACY

89.9%
MACRO F1 SCORE

100%
RETRIEVAL RECALL@3

0.0%
REFUTATION FALSE NEG

VERIFIED FEVER CONFUSION MATRIX (PAPER_DEV)

Actual \ Predicted	Contradicted	Supported	Total
Contradicted (Refuted)	4 (True Pos)	0 (False Neg)	4
Supported (Entailed)	1 (False Pos)	5 (True Pos)	6

- **Refutation Safety: 100% of contradicted claims (4/4) correctly flagged (0.0% False Negative Rate).**
- Retrieval Precision: 100% of ground-truth evidence passages retrieved in Top-3.
- Macro Metrics: Precision = 90.0%, Recall = 91.7%, Macro F1 = 89.9%.

END-TO-END LATENCY BREAKDOWN (WARM INFERENCE)

Pipeline Stage	Execution Time	Latency Share
1. Claim Extraction	1.9 ms	0.6 %
2. Hybrid Retrieval	8.4 ms	2.8 %
3. Cross-Encoder NLI	285.2 ms	96.3 %
4. Aggregation & Scoring	0.8 ms	0.3 %

TOTAL WARM LATENCY: 296.3 ms / claim (Interactive Sub-300ms Performance)

Note: Large-scale 10,000-sample HaluEval evaluation is actively running; results shown reflect verified local FEVER paper_dev runs.

Conclusion & Future Scope

Summary of verified project capabilities, acknowledged limitations, and future extensions.

VERIFIED PROJECT CAPABILITIES

✓ **Claim-Level Decomposition:**
Deconstructs compound sentences into atomic propositions with exact character-span offsets, enabling pinpoint user interface highlighting.

✓ **Hybrid FAISS + BM25 Retrieval:**
Fuses 384-dimensional dense semantic vectors with lexical keyword matching, achieving 100% Recall@3 on tested reference benchmarks.

✓ **Calibrated Cross-Encoder NLI:**
DeBERTa-v3 joint attention arbitrates multi-passage evidence into calibrated verdicts, achieving 90.0% accuracy on FEVER with 0.0% false negatives.

✓ **Closed-Loop Reverification:**
Eliminates secondary hallucinations through evidence-grounded prompt constraints followed by mandatory reverification.

ACKNOWLEDGED LIMITATIONS & FUTURE SCOPE

► **Single-Hop Reasoning Boundary:**
Cross-encoder evaluates individual passages; cannot chain facts across distant chapters. Roadmap: Graph-RAG multi-hop reasoning.

► **Static Decision Thresholds:**
Thresholds (0.55 / 0.50) are globally configured. Roadmap: Implement dynamic conformal prediction intervals per domain.

► **Inference Acceleration:**
Current 296ms latency is suitable for desktop use, but ONNX INT8 runtime quantization will push verification below 100ms for edge deployment.

► **Domain-Adaptive Fine-Tuning:**
Continual pre-training of DeBERTa-v3 on specialized clinical, legal, and financial hallucination corpora.

References

Key peer-reviewed literature in hallucination detection, natural language inference, and dense retrieval.

CORE RESEARCH LITERATURE

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DEFENSE CONCLUSION // OPEN DISCUSSION

Questions & Answers

TruthGuard: Explainable Hallucination Detection & Verification Framework

"Fluency cannot replace evidence. TruthGuard demonstrates that real-time, explainable hallucination detection and closed-loop self-correction are achievable on consumer hardware in under 300 milliseconds."

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